Stylometric Analysis of AI-Generated Text: Lexical Fidelity vs. Structural Limitations

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ABSTRACT

This exploratory study examines whether AI-generated imitations are stylometrically closer to their intended source texts than to unrelated texts. We analyze 18 imitations of three King James Bible chapters produced by three generative models under two prompt conditions. Each text is represented using lexical, KJV-specific, function-word, and sentence-level structural features, and imitation fidelity is evaluated through a matching versus non-matching distance framework. AI-generated texts were generally closer to their intended source chapters than to unrelated chapters, and this pattern remained after z-score standardization. Lexical and KJV-specific markers were reproduced relatively closely, while sentence-level structural properties showed greater variability, though part of the raw difference reflected feature scale. These findings provide preliminary evidence that stylistic imitation is uneven across linguistic dimensions. Because the dataset is small and limited to a distinctive historical corpus, the results should not be interpreted as evidence for a general-purpose AI-text detection method.

KEYWORDS

Stylometry; AI-generated Text; Authorship Attribution; Stylistic Imitation; Feature Analysis

INTRODUCTION

The question of whether a text was written by a human or a machine has become an important practical issue in today's information environment. In education, this issue relates to academic integrity. Teachers may ask whether a student wrote an essay or whether AI produced it. In journalism and scholarly publishing, the issue relates to credibility and transparency. In legal and professional settings, authorship also matters because it carries accountability. These concerns have grown with the rise of large language models. These models can produce fluent text in many different styles and registers. As a result, surface reading alone is often no longer enough to tell whether a text is human-written or AI-generated. This makes systematic and feature-based methods of authenticity assessment more important.

Authorship attribution offers an established foundation for studying this problem in information science, linguistics, and digital humanities. Stylometry approaches authorship by examining measurable patterns in text. It treats writing style as something that can be quantified, rather than judged only by impression. Early studies showed that simple features, such as word length and vocabulary use, could distinguish between authors. De Morgan (1882) suggested this idea, and later work developed it further through computational methods (Holmes, 1998). Over time, stylometric research expanded to include function word frequencies, lexical diversity, and structural features. These developments made it possible to represent texts as points in a multidimensional feature space. In this framework, stylistic similarity can be measured through distance-based comparison. Authorship can then be inferred from proximity in that space.

Generative AI complicates this framework in an important way. Most practical concerns about AI-generated text are framed as a binary problem. The question is usually whether a text is human-written or machine-generated. This view has shaped much of the existing work on AI detection. However, a different and less studied problem appears when AI is used not just to generate text, but to imitate a specific style. In practice, users may ask AI to copy their own writing, imitate another person's style, or produce text in a historically specific register. In these cases, the key question is no longer simple authorship. The issue becomes stylistic conditioning. The question is whether the generated text matches the distributional patterns of a specific source. A text may look convincing to human readers while still failing to reproduce the deeper structure of its intended model. For this reason, understanding where imitation succeeds and where it fails is a separate problem from both authorship attribution and AI detection.

Studying this problem requires moving beyond surface plausibility and examining style at the feature level. This study analyzes AI-generated imitations of selected chapters from the King James Bible (KJV), a corpus with a distinctive but internally varied style. Each text is represented as a vector of stylometric features. These features capture lexical, structural, and KJV-specific properties. Stylometric similarity is measured through feature-based distances in a matching versus non-matching design. Each AI-generated text is compared both to its source chapter and to unrelated chapters. This design makes it possible to test source-conditioned stylistic alignment directly. Traditional authorship attribution asks whether two texts share the same author. This study asks instead whether a generated text is closer to its intended source than to other possible stylistic targets.

This exploratory study makes three focused contributions. First, it adapts a matching versus non-matching distance framework to evaluate source-conditioned AI imitation in stylometric feature space. Second, it provides a feature-

level comparison of lexical, KJV-specific, function-word, and sentence-level structural properties. Third, it identifies a preliminary asymmetry in imitation fidelity: under the conditions examined, generated texts approximate lexical markers more closely than some sentence-level structural distributions. These findings should be interpreted as exploratory patterns from a small and stylistically specialized corpus rather than as general conclusions about AI-generated text.

RELATED WORK

Stylometry and Feature-Based Representation

Stylometry models writing style using measurable linguistic features. Early studies showed that simple features, such as word length and vocabulary usage, can help distinguish between authors (Juola, 2006). The development of stylometry as a scholarly discipline has been closely tied to advances in computational methods, enabling increasingly systematic analysis of textual style across literary and historical corpora (Holmes, 1998). Later research found that features like function word frequencies, sentence length, and lexical diversity tend to remain consistent within an author's work but vary across different authors (Stamatatos, 2009). These features are commonly used to represent texts as vectors in a multidimensional feature space for authorship analysis. More advanced methods include character n-grams and syntactic patterns, which improve robustness and capture deeper stylistic variation (Koppel et al., 2009). Within this framework, stylistic similarity is measured by comparing patterns across these features quantitatively.

Distance-Based Methods and Feature Space

Distance-based methods provide a way to understand stylistic similarity in geometric terms. One of the most well-known approaches, Burrows' Delta, measures the standardized distance between texts based on their feature values, assuming that texts by the same author are closer together than those by different authors (Burrows, 2002). Similar methods have been widely used in literary and biblical studies, where stylistic closeness is treated as evidence of shared authorship (Savoy, 2019). Instead of only assigning texts to categories, these approaches focus on how texts are positioned relative to each other in stylometric space, allowing for a more detailed analysis of stylistic relationships.

AI-Generated Text and Detection

The emergence of generative AI challenges traditional assumptions in stylometry. Large language models can produce text that closely resembles human writing, including historically specific styles. Recent research has focused on detecting AI-generated text using statistical methods, such as likelihood-based detection and perturbation-based approaches like DetectGPT (Gehrmann et al., 2019; Mitchell et al., 2023). Subsequent work has expanded this line of investigation, including benchmark environments for evaluating detection methods across multiple generators (Uchendu et al., 2021), watermarking-based approaches (Kirchenbauer et al., 2023), and analyses questioning the reliability and fairness of AI text detectors across different populations (Liang et al., 2023; Sadasivan et al., 2024). However, these studies mainly treat the problem as a binary classification task, focusing on distinguishing human from machine-generated text rather than examining how the stylistic structure of imitation is organized internally.

Gap: Stylometric Positioning of AI Imitation

Despite the growing body of detection research reviewed above, relatively little work has examined how AI-generated texts behave within stylometric feature space. Existing studies do not examine which stylistic features AI reproduces well and which it fails to reproduce. As a result, it is still unclear whether AI-generated texts are actually closer to their intended targets than to unrelated texts, and which features are reproduced accurately versus distorted. This gap motivates a distance-based and feature-level approach to AI-generated text. Instead of treating stylistic imitation as a simple classification result, this approach studies it as a continuous and multidimensional pattern.

RESEARCH METHOD

Data and Study Design

This study examines whether AI-generated texts are closer to their intended source texts than to unrelated texts in stylometric feature space. The dataset includes selected chapters from the King James Version (KJV) of the New Testament and AI-generated imitations produced by several models, including ChatGPT 5.4, Claude Sonnet 4.6, and Gemini 3.1 Pro. The KJV was selected because it has a distinctive and historically stable style. This style includes archaic pronouns, characteristic verb forms, and consistent discourse patterns. At the same time, different chapters still show measurable stylistic variation.

Three chapters were selected as source texts: John 1, Luke 15, and Matthew 5. These chapters were deliberately selected to represent distinct rhetorical modes within the KJV, as summarized in Table 1. This selection ensures that the study captures variation not only in vocabulary but also in sentence-level structure, providing a stronger basis for testing whether AI models can reproduce different rhetorical patterns within the same stylistic register. For each chapter, one AI-generated text was produced under each prompt condition (old and new) by each model, yielding a

total of 18 AI-generated texts (3 chapters × 3 models × 2 prompts × 1 sample), allowing for controlled comparison of prompt effects. The use of a single sample per condition reflects the exploratory scope of the study, which prioritizes breadth of comparison across models, chapters, and prompt conditions over within-condition replication. This design supports hypothesis testing at the group level but does not permit estimation of within-condition variability. Individual results, particularly extreme values, should therefore be interpreted as illustrative rather than as stable estimates of model behavior. The two prompt conditions differed intentionally. The old prompt provided only minimal guidance, asking models to imitate the KJV style without specifying which features to prioritize. The new prompt was more detailed: it directed models to focus first on sentence-level structural patterns, including sentence length distribution, rhythmic pacing, and clause-level structures, while treating lexical features such as archaic vocabulary as secondary. This difference was designed to test whether explicit structural guidance could improve imitation in the dimensions where AI-generated texts were expected to perform less well. The experimental design distinguishes between matching pairs (AI-generated text vs. its source chapter) and non-matching pairs (AI-generated text vs. other chapters). This framework enables direct evaluation of whether AI imitation is source-conditioned rather than merely genre-level.

Chapter	Rhetorical Mode	Words	Sentences	Avg Sent. Length	TTR	Key Structural Property
John 1	Theological prologue	1006	60	16.77	0.27	Short declarative sentences, abstract language
Luke 15	Narrative (parables)	726	29	25.03	0.36	Extended storytelling, complex clause embedding
Matthew 5	Didactic discourse	1082	42	25.76	0.29	Parallel structure, formulaic repetition

Table 1. Source Chapter Characteristics

Stylometric Feature Extraction

Each text is represented as a feature vector in a *d*-dimensional stylometric space:

$$X = (x_1, x_2, \ldots, x_d)$$

where each dimension corresponds to a stylometric feature. The feature set includes:

- Lexical features: type-token ratio (TTR), average word length
- Stylistic (KJV-specific) features: archaic pronoun ratio, biblical verb ratio, discourse marker ratio
- Function word features: frequencies of function words (e.g., *and, of*)
- Structural features: average sentence length, sentence length standard deviation, sentence length quartiles (Q25, median, Q75), comma rate, semicolon rate

Frequency-based lexical, stylistic, function-word, and punctuation features were normalized by total word count, whereas sentence-length features were computed from the distribution of sentence lengths within each text:

$$f_i = \frac{count_i}{\text{total words}}$$

This representation embeds both human-authored and AI-generated texts in a common stylometric space.

Distance-Based Comparison Framework

Stylometric similarity is measured using Euclidean distance between feature vectors:

$$D(\mathrm{x}, \mathrm{y}) = \sqrt{\sum_{i=1}^{d} (x_i - y_i)^2}$$

In this space, smaller distances indicate greater stylistic similarity.

The analysis is based on a matching versus non-matching framework. For each AI-generated text x_{AI}, its distance to the source chapter x_{source} is defined as:

$$D_{match} = D\,(\mathrm{x}_{AI}, \mathrm{x}_{source})$$

and its distance to other chapters is defined as:

$$D_{non} = D\,(\mathrm{x}_{AI}, \mathrm{x}_{other})$$

The core hypothesis is that:

$$D_{match} < D_{non}$$

If this condition holds consistently, it indicates that AI-generated texts are positioned closer to their intended sources, demonstrating source-conditioned stylistic imitation.

Distances are further grouped by prompt condition, model, and chapter to examine variation in imitation quality across different experimental settings. Features were not standardized prior to distance computation in the primary analysis because the raw values preserve the interpretability of absolute deviations. However, the feature set combines variables measured on substantially different scales. Sentence-length variables are measured in numbers of words, whereas many lexical and stylistic variables are proportions bounded between 0 and 1. Consequently, raw structural and lexical distances are not directly comparable in magnitude. Raw structural-to-lexical ratios are therefore treated only as descriptive summaries rather than as precise estimates of relative imitation difficulty. In addition, the structural feature group contains several related sentence-length measures, including the mean, standard deviation, Q25, median, and Q75, which may give sentence-length variation greater effective weight. A supplementary z-score standardized analysis is used to evaluate whether the main matching versus non-matching result remains after controlling for feature scale.

Statistical and Feature-Level Analysis

To evaluate whether matching distances are significantly smaller than non-matching distances, an independent-samples t-test is conducted. The test statistic is defined as:

$$t = \frac{\mu_{match} - \mu_{non}}{\sqrt{\frac{s^2_{match}}{n_{match}} + \frac{s^2_{non}}{n_{non}}}}$$

where μ denotes the group mean, s^2 the sample variance, and n the sample size of each group. A significant result indicates systematic separation between matching and non-matching pairs.

To further analyze imitation fidelity, feature-level deviation is computed as:

$$\Delta f_i = \left| f_i^{AI} - f_i^{human} \right|$$

and averaged across samples:

$$\overline{\Delta f_i} = \frac{1}{n} \sum_{j=1}^{n} \left| f_{i,j}^{AI} - f_{i,j}^{human} \right|$$

This allows identification of which features contribute most to stylometric differences. To further examine the source of stylometric deviation, distances were decomposed into two sub-components. The structural distance was computed using sentence-level features (average sentence length, standard deviation, Q25, median, Q75, comma rate, and semicolon rate), while the lexical distance was computed using word-level features (TTR, average word length, archaic pronoun ratio, biblical verb ratio, discourse marker ratio, and frequencies of *and* and *of*). This decomposition allows for direct comparison of how faithfully AI-generated texts reproduce each dimension of style.

Visualization

Results are visualized using bar plots, boxplots, and heatmaps to illustrate distance distributions, pairwise similarities, and feature-level deviations. These visualizations highlight clustering patterns, differences across prompts and models, and the relative difficulty of reproducing different stylistic dimensions.

RESULTS

Validation of Stylometric Distance Principle

The central hypothesis of this study holds that AI-generated texts are positioned closer to their intended source chapters than to unrelated chapters in stylometric feature space, such that $D_{match} < D_{non}$. This prediction was tested across all 18 AI-generated texts and three human source chapters, yielding 18 matching pairs and 36 non-matching pairs.

Matching distances were descriptively lower than non-matching distances. The mean matching distance was 7.99 (*SD* = 8.18), compared with a mean non-matching distance of 16.57 (*SD* = 8.31). An independent-samples t-test indicated a difference between the two sets of distances (t = -3.61, $p < 0.001$). However, this test should be interpreted cautiously because the non-matching observations are not fully independent: each AI-generated text appears in two non-matching comparisons. The test therefore provides descriptive rather than definitive inferential

evidence. Nevertheless, the same directional pattern appeared across all model and prompt conditions and remained present in the z-score standardized analysis, providing convergent support for source-conditioned stylistic proximity. The same directional pattern was observed across both prompt conditions and all three models, although the magnitude of separation varied substantially across individual texts. Taken together, these descriptive and standardized results support the usefulness of the distance-based framework and suggest source-conditioned stylistic proximity within the present dataset. The models appear to reproduce more than a generic KJV register, although this interpretation should be tested with a larger and more diverse sample. Figure 1 illustrates this separation across all models and prompt conditions combined.

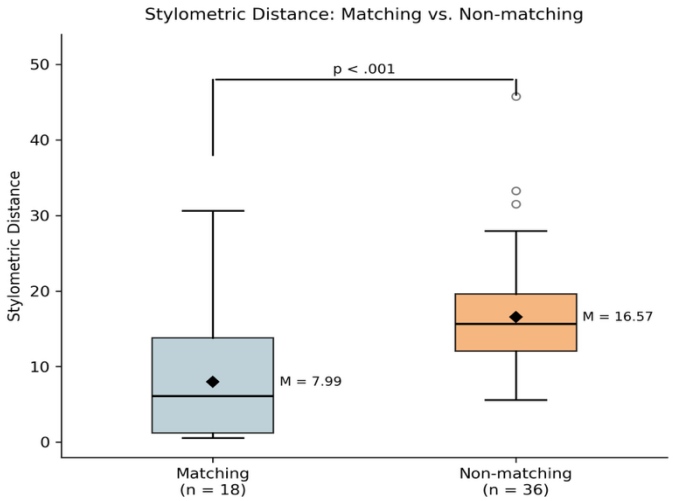


Figure 1. Distribution of Stylometric Distances: Matching vs. Non-matching Pairs

Prompt-Level Comparison

To examine whether prompt formulation influenced imitation quality, matching and non-matching distances were compared separately under the old and new prompt conditions. Under the old prompt, matching distances (M = 7.35, SD = 6.06) were substantially lower than non-matching distances (M = 14.74, SD = 6.29). The new prompt produced the same directional pattern, with matching distances (M = 8.64, SD = 10.22) remaining well below non-matching distances (M = 18.40, SD = 9.77). Mean matching distances were similar across the two prompt conditions, and the new prompt did not produce a consistent overall improvement. Given that each model–chapter–prompt condition contains only one generated text, these values are interpreted descriptively rather than as a stable estimate of prompt effectiveness. The prompt manipulation remains diagnostically useful because it reveals differences in how models respond to structural guidance.

Pairwise Stylometric Structure

Figure 2 presents the full pairwise stylometric distance matrix across all 21 texts. Human chapters cluster with their corresponding AI imitations along the diagonal blocks, confirming source-conditioned proximity. The degree of clustering varies by model: Gemini 3.1 Pro and ChatGPT 5.4 imitations appear consistently close to their source chapters, while Claude Sonnet 4.6 imitations showed greater variability. Matching distances ranged from 0.55 (John 1, ChatGPT, old prompt) to 30.62 (Luke 15, Claude, new prompt), illustrating that source-conditioned proximity varies considerably across individual combinations. The extreme value of 30.62 should be interpreted as an illustrative case rather than a stable estimate.

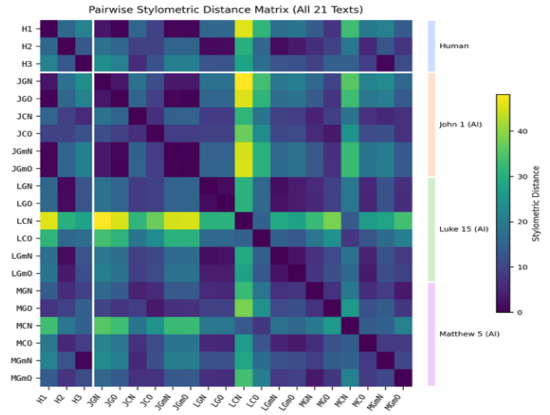


Figure 2. Pairwise Stylometric Distance Matrix Across All 21 Texts

Model-Level Performance

Substantial differences in imitation fidelity were observed across the three models. Averaged across all chapters and prompt conditions, Gemini 3.1 Pro produced the lowest mean matching distance (M = 3.07, SD = 4.03), followed by ChatGPT 5.4 (M = 5.00, SD = 5.77). Claude Sonnet 4.6 produced substantially larger matching distances (M = 15.92, SD = 7.90), approximately three times those of the other two models. Given that each model-chapter-prompt combination is represented by a single sample, these model-level differences should be interpreted as suggestive patterns rather than definitive performance rankings. Figure 3 presents matching distances by model and prompt condition.

Prompt condition interacted with model performance in a notable pattern. For Gemini 3.1 Pro, the new prompt produced substantially lower matching distances (M = 1.07) than the old prompt (M = 5.07), suggesting that explicit structural guidance benefited this model. ChatGPT 5.4 showed a similar but smaller effect (new: M = 4.44; old: M = 5.55). Claude Sonnet 4.6, however, showed the opposite pattern: matching distances increased under the new prompt (M = 20.41) relative to the old (M = 11.43). This suggests that the effect of detailed structural prompting on imitation fidelity is model-dependent: for some models, additional structural constraints may increase rather than reduce stylometric divergence from the source.

These results suggest that model identity may be a stronger determinant of imitation fidelity than prompt condition, and that the relationship between prompt specificity and imitation fidelity is model-dependent rather than uniform.

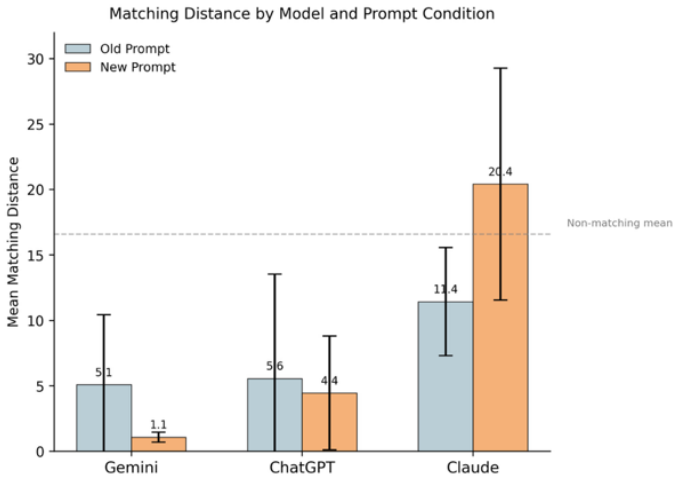


Figure 3. Mean Matching Distance by Model and Prompt Condition

Chapter-Level Variation

Imitation difficulty varied across source chapters (Table 2). John 1 was most faithfully reproduced (M = 5.00, SD = 6.21), followed by Luke 15 (M = 9.04, SD = 12.06) and Matthew 5 (M = 9.94, SD = 5.17). These differences align with rhetorical properties: John 1's short declarative sentences were comparatively accessible, while Luke 15's extended narrative structure and Matthew 5's mix of formulaic and doctrinal patterns proved harder to reproduce. The large deviation for Claude's Luke 15 imitation under the new prompt (D = 30.62) illustrates how structural

prompting may interact poorly with certain combinations, though this single-sample result should not be generalized. Prompt condition interacted with chapter difficulty inconsistently (Table 2).

Chapter	Old Prompt	New Prompt	Overall M	Overall SD
John 1	3.55	6.45	5.00	6.21
Luke 15	6.99	11.09	9.04	12.06
Matthew 5	11.51	8.37	9.94	5.17

Table 2. Mean Matching Distance by Chapter and Prompt Condition

Feature-Level Analysis

Figure 4 shows the mean absolute deviation for each feature across all matching pairs, with features grouped by type. The results reveal a clear difference between structural and lexical features. Structural features at the sentence level showed the largest deviations. Sentence length Q75 had the highest mean deviation ($M = 5.46$), followed by average sentence length ($M = 2.88$), sentence length median ($M = 2.47$), and sentence length Q25 ($M = 1.89$). Sentence length standard deviation also showed a relatively large deviation ($M = 1.59$). These results indicate that AI-generated texts differed most from human texts in sentence-level structure.

By contrast, lexical features remained much more stable across models. Type-token ratio ($M = 0.031$) and average word length ($M = 0.104$) showed only small deviations. KJV-specific features were also reproduced with relatively high consistency, including archaic pronoun ratio ($M = 0.007$), biblical verb ratio ($M = 0.002$), and discourse marker ratio ($M = 0.005$). The frequencies of function words such as and of were also closely matched.

This pattern appeared across all three models, although the size of the difference varied. Claude Sonnet 4.6 showed the largest structural deviations. Its sentence length Q75 differed from the human source by an average of 10.08 words, compared with 3.88 for ChatGPT 5.4 and 2.71 for Gemini 3.1 Pro. In lexical features, however, all three models performed similarly well. These findings suggest that current generative models can reproduce lexical style quite effectively, but they still struggle with structural imitation, especially in matching sentence length distributions.

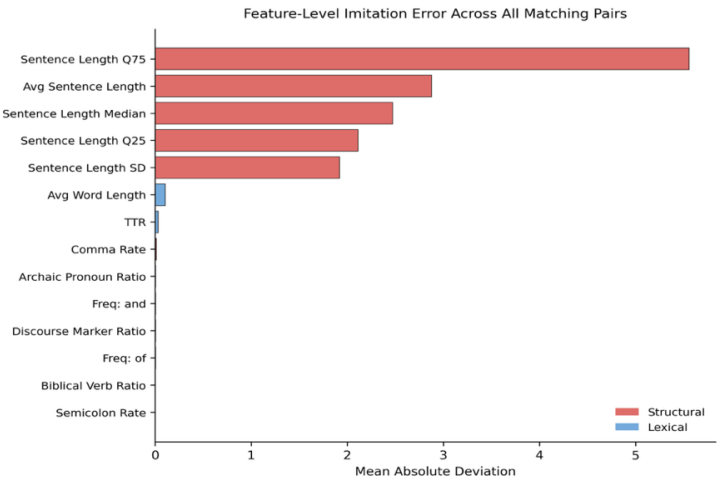


Figure 4. Mean Absolute Deviation per Feature Across All Matching Pairs

Prompt × Feature Interaction

Comparing feature-level deviations across prompt conditions reveals an unexpected asymmetry. The new prompt explicitly asked models to focus on structural imitation while treating lexical features as secondary. If effective, structural deviations should have decreased and lexical deviations should have stayed similar or increased. Table 3 summarizes the results. The prompt reduced lexical deviations, even though lexical style was described as secondary. Structural features, however, did not improve and in several cases became less accurate under the new prompt. This disconnect between prompt guidance and generation outcomes suggests that structural imitation difficulty is not simply a matter of where the prompt directs model attention. Prompting can influence surface-level features, but it appears less effective at controlling sentence-level distributional patterns under the conditions tested.

Feature	Type	Old Prompt	New Prompt	Direction
Type-token ratio	Lexical	0.038	0.024	↓Improved
Avg word length	Lexical	0.121	0.087	↓Improved
Avg sentence length	Structural	2.26	3.49	↑Worsened
Sentence length Q25	Structural	1.44	2.78	↑Worsened

Table 3. Effect of Prompt Condition on Feature-Level Deviation

Structural vs Lexical Distance

The distance decomposition analysis shows a consistent pattern across all conditions. Figure 5 separates structural and lexical distances for matching and non-matching pairs under both prompt conditions.

In the main unstandardized analysis, structural features contributed most of the observed distance. Under the old prompt, the mean structural distance for matching pairs was 7.35, while the lexical distance was 0.13. In raw terms, this gives a ratio of about 55:1. Under the new prompt, structural distance was 8.64 and lexical distance was 0.09, which gives a ratio of about 92:1. Non-matching pairs showed the same pattern, but at larger absolute values. These ratios should be interpreted carefully, because sentence-level features are measured in word counts while lexical features are measured as proportions. Part of the gap therefore comes from scale differences, not only from differences in imitation difficulty.

To test whether the same pattern remains after controlling for scale, a supplementary analysis used z-score standardized features. Figure 5 reports these results. After standardization, the separation between matching pairs (M = 2.72, SD = 1.77) and non-matching pairs (M = 5.74, SD = 0.98) became even stronger ($t = -6.75$, $p < 0.001$), which supports the robustness of source-conditioned proximity. In the standardized space, mean structural distance for matching pairs (M = 1.73) and mean lexical distance (M = 1.94) were similar in size. This suggests that the large raw ratios were mostly caused by scale differences. Even so, structural distances remained consistently larger for non-matching than for matching pairs across all conditions. The ordering across models also stayed the same after standardization, with Gemini 3.1 Pro (M = 1.36) performing better than ChatGPT 5.4 (M = 2.30) and Claude Sonnet 4.6 (M = 4.50).

Taken together, these results support the same core conclusion. AI-generated texts are consistently closer to their source chapters than to unrelated chapters, and this result holds in both unstandardized and standardized analyses. The claim that structural features dominate in raw magnitude needs to be qualified because of scale. Even so, the consistent lexical–structural pattern across models, chapters, and prompt conditions remains the main empirical finding.

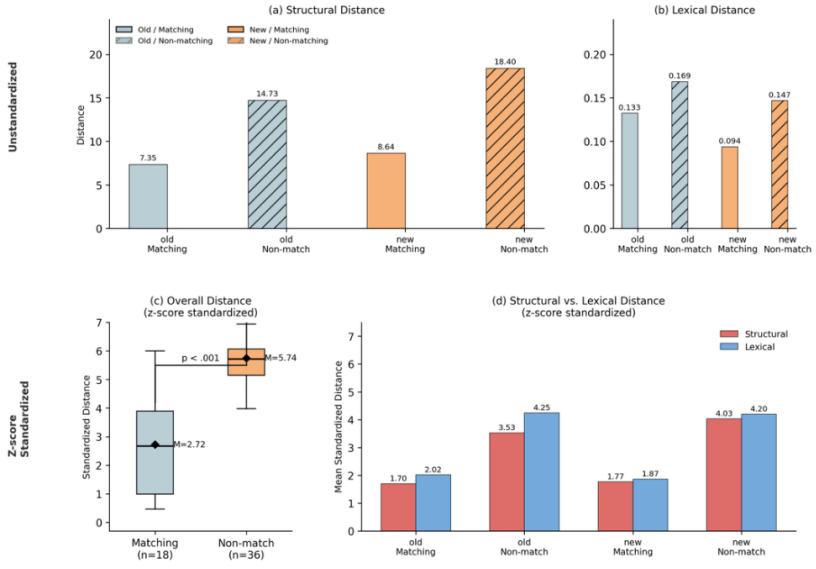


Figure 5. **Distance Decomposition: Unstandardized (top row) and Z-score Standardized (bottom row). (a) Structural distance by prompt condition and comparison type; (b) lexical distance by prompt condition and comparison type; (c) overall matching vs. non-matching separation after standardization ($t = -6.75$, $p < 0.001$); (d) structural vs. lexical distance after standardization.**

DISCUSSION

This study examined whether AI-generated imitations of King James Bible chapters are positioned closer to their source texts than to unrelated texts in stylometric feature space, and which features drive the observed distances. The results support the core hypothesis while revealing a systematic asymmetry that has broader implications for how stylistic imitation should be understood and measured.

Stylometric Theory under AI

The matching versus non-matching framework introduced in this study provides a way to evaluate AI imitation that goes beyond binary classification. Rather than asking whether a text is human or machine-generated, it asks whether a generated text is stylometrically conditioned on its intended source. The significant separation between matching and non-matching distances ($t = -3.61$, $p < 0.001$), further confirmed by the supplementary standardized analysis ($t = -6.75$, $p < 0.001$), demonstrates that this conditioning is real and measurable: AI-generated texts are not simply reproducing a generic KJV register but are producing outputs that are demonstrably closer to specific target chapters. The primary contribution is not methodological but empirical: the demonstration that AI imitation is source-conditioned yet structurally incomplete. This finding extends the distance-based tradition in stylometry, represented by approaches such as Burrows' Delta and intertextual distance (Burrows, 2002; Savoy, 2019), into the domain of generative AI evaluation.

Nature of AI Imitation

The most important theoretical finding concerns how imitation fails internally. Across all models and prompt conditions, AI-generated texts showed strong lexical alignment with their source chapters, reproducing archaic vocabulary, function word frequencies, and other word-level features with very little deviation. In the unstandardized analysis, structural features showed deviations 55 to 92 times larger than lexical deviations in absolute terms, though the supplementary standardized analysis revealed that this gap largely reflects scale differences. After standardization, structural and lexical distances were comparable in magnitude, but both feature types showed consistent directional separation between matching and non-matching pairs. This pattern suggests that current generative models operate differently across these two levels of style. One possible explanation is that lexical patterns are well represented in training data and can be reproduced through the token-level prediction process used by large language models. Sentence-level structural features, especially the distribution of sentence lengths, may not be captured as effectively, as they require coordination across longer spans of text. Testing this explanation directly would require controlled experiments with model internals, which is beyond the scope of the present study.

The prompt manipulation provides diagnostic support for this interpretation, not through a significant global effect, but by revealing differential model responses across feature types. The new prompt explicitly asked models to focus on sentence structure while placing less emphasis on lexical features. Models responded partly at the lexical level, showing lower deviations in type-token ratio and average word length. Structural deviations, however, did not improve and in several cases became larger. This disconnect between prompt guidance and generation outcomes suggests that structural imitation difficulty is not simply a matter of where the prompt directs model attention but is consistent with a limitation in how current generative models coordinate text production at the sentence level.

The chapter-level results add an important dimension to this interpretation. The three source chapters represent different rhetorical modes: theological exposition (John 1), narrative storytelling (Luke 15), and didactic discourse (Matthew 5). Imitation difficulty was not uniform across these modes. Narrative text, with its extended sentence sequences and sustained clause-level complexity, proved hardest to reproduce. This suggests that the structural limitation is not only a matter of sentence-level statistics in isolation but also relates to the sequential coherence demands of different discourse types. Models that generate text token by token may find it especially difficult to maintain consistent structural patterning across the longer spans required by narrative prose.

Hierarchy of Features

These findings suggest a hierarchy of stylometric features based on how easily generative models can reproduce them. Surface lexical features, such as vocabulary choice, archaic markers, and function word frequencies, are the most easily reproduced. Within this group, KJV-specific features such as archaic pronoun ratios and biblical verb ratios were among the most faithfully matched, suggesting that stylistically distinctive vocabulary markers are particularly accessible when the target register is well represented in training data. Structural features form a deeper and more resistant level. In particular, sentence length distributions and their variation showed greater variability across all conditions in both the unstandardized and standardized analyses, supporting their value as diagnostic signals even after controlling for feature scale.

For AI-detection research, this pattern provides a tentative feature-selection hypothesis rather than evidence for a deployable detection method. Lexical features may become less informative when a model is explicitly prompted to imitate a distinctive register, while sentence-level distributions may deserve closer examination. However, the

usefulness of these features for detection cannot be established from the present dataset and should be evaluated across larger samples, additional genres, and multiple outputs per condition. This extends Holmes' (1998) argument that durable stylometric signals lie in structural properties and aligns with Kenny's (1986) structural analysis of biblical texts.

Implications

Implications for Stylometric Research and AI Detection

These findings show that feature selection matters when evaluating AI imitation. Studies focusing mainly on lexical features may overestimate AI–human similarity. Sentence-level structural properties, especially sentence-length distributions, warrant further evaluation as candidate features in studies of stylistically prompted AI text. The present results do not establish their reliability in operational detection systems.

Implications for Information Science and Policy

The results also suggest that authenticity assessment should not rely exclusively on whether a text reproduces recognizable vocabulary or appears stylistically plausible. Nevertheless, the present evidence is limited to a small set of KJV-based imitations and does not support direct policy recommendations. The findings are better understood as motivating further research on how lexical and structural evidence can be combined in document-provenance assessment.

Implications for Generative AI Research

The results suggest a gap between what models can be prompted to do and what they reliably produce. Structural fidelity varied across models, and explicit structural prompting did not improve sentence-level accuracy. Longitudinal stylometric evaluation offers a useful direction for future research.

Limitation and Future Work

Several limitations should be noted. First, the dataset is small (18 AI-generated texts from three chapters, three models, two prompts). Although 54 pairwise comparisons support the main hypothesis, statistical power for detailed sub-comparisons is limited. The consistent directionality across all conditions provides convergent support. Second, all source texts come from a single register (KJV). The consistency of the pattern across rhetorically distinct chapters suggests the finding is not narrowly tied to one textual form, but future work should test modern prose, academic writing, and legal language. Third, the primary analysis used unstandardized features, and raw structural-to-lexical ratios (55:1, 92:1) overstate the asymmetry. The supplementary standardized analysis confirmed the core finding with a stronger effect ($t = -6.75$, $p < 0.001$), though structural and lexical distances became comparable in magnitude after standardization. Additionally, the five sentence-length features included in the structural group (mean, standard deviation, Q25, median, and Q75) are likely to be mutually correlated, which may further inflate structural distances by effectively weighting sentence-length variation multiple times. Future work could address this through dimensionality reduction or by selecting a smaller set of orthogonal structural features. Fourth, each condition includes one sample, limiting separation of stable behavior from random variation. Finally, models change rapidly, making longitudinal evaluation important.

CONCLUSION

This exploratory study examined whether AI-generated imitations of selected King James Bible chapters are stylometrically closer to their intended source chapters than to unrelated chapters. Using a matching versus non-matching design, the analysis found that generated texts were generally positioned closer to their target chapters. This directional pattern remained after z-score standardization, suggesting that it was not solely an artifact of differences in feature scale. Feature-level results indicate that imitation fidelity was uneven across linguistic dimensions. Lexical and KJV-specific markers were reproduced relatively closely, whereas sentence-level structural properties showed greater variability. At the same time, the large structural-to-lexical differences observed in raw distances were partly caused by measurement scale and the inclusion of several related sentence-length features. The evidence therefore supports a cautious conclusion: source-conditioned AI imitation is measurable, but it is not uniformly faithful across feature groups. These findings do not establish a general detector of AI-generated text or a definitive ranking of generative models. Rather, they demonstrate the usefulness of feature-level stylometric analysis for studying how AI systems imitate specific source texts. Future studies should include multiple outputs per condition, more diverse textual registers, and statistical procedures that explicitly account for dependent comparisons.

GENERATIVE AI USE

We employed Claude Sonnet 4.6 (Anthropic) for the following purposes: Python code generation for data analysis and language refinement during the drafting of this manuscript. All statistical results, interpretations, and conclusions were verified by the authors against the original data. All AI-generated text was reviewed, edited, and revised by the authors for accuracy, consistency, and alignment with the research findings. The authors assume all responsibility for the content of this submission.

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